STACKS, DISCRIMINANTS: LEMMA-NOETHER-DIFFERENT-FLAT-QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1484-1499 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1501-1522, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 28 of queue_discriminant.json (pb=986, stmt 1484-1499, proof 1501-1522).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BVS: sources/stacks-discriminant/pinned-original-source.tex, source lines 1428–1432. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a flat quasi-finite homomorphism of Noetherian rings. The diagram*

$$\begin{array}{ccc} J & \longrightarrow & \mathrm{Hom}_B(\omega_{B/A}, B) \\ & {\scriptstyle\mu}\searrow \quad \swarrow{\scriptstyle\varphi\mapsto\varphi(\tau_{B/A})} & \\ & B & \end{array}$$

commutes where the horizontal arrow is the isomorphism of Lemma 0BVS. Hence the Noether different of B over A is the image of the map $\mathrm{Hom}_B(\omega_{B/A}, B) \to B$.

Proof. Exactly as in the proof of Lemma 0BVS this reduces to the case of a finite free map $A \to B$. In this case $\tau_{B/A} = \mathrm{Trace}_{B/A}$. Choose a basis $b_1, \ldots, b_n$ of B as an A-module. Let $\xi = \sum b_i \otimes c_i \in J$. Then $\mu(\xi) = \sum b_i c_i$. On the other hand, the image of ξ in $\mathrm{Hom}_B(\omega_{B/A}, B)$ sends $\mathrm{Trace}_{B/A}$ to $\sum \mathrm{Trace}_{B/A}(b_i)c_i$. Thus we have to show

$$\sum b_i c_i = \sum \mathrm{Trace}_{B/A}(b_i)c_i$$

when $\xi = \sum b_i \otimes c_i \in J$. Write $b_i b_j = \sum_k a_{ij}^k b_k$ for some $a_{ij}^k \in A$. Then the right hand side is $\sum_{i,j} a_{ij}^j c_i$. On the other hand, $\xi \in J$ implies

$$(b_j \otimes 1)(\sum\nolimits_i b_i \otimes c_i) = (1 \otimes b_j)(\sum\nolimits_i b_i \otimes c_i)$$

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which implies that $b_j c_i = \sum_k a^i_{jk} c_k$. Thus the left hand side is $\sum_{i,j} a^i_{ij} c_j$. Since $a^k_{ij} = a^k_{ji}$ the equality holds. $\square$